



EE 354/CE 361/MATH 310 -
Introduction to Probability
and Statistics
Fall 2025



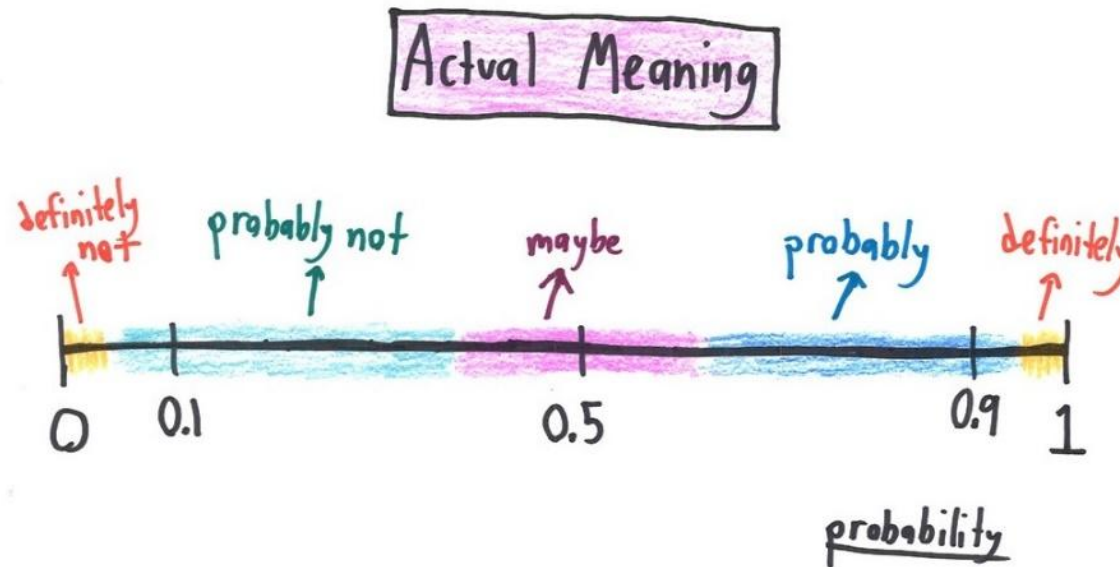


EE 354/CE 361/MATH 310 L2 section (Fall 2025)

Introduction to Probability and Statistics -

Aamir Hasan

Week 3 - Lecture No 04 – 1st September 2025



Announcements!

1. Lectures up to No 03 posted
2. Quiz No 01 – next class on September 03, 2025 (Wednesday) from Unit No 01

Agenda for today

1. Probability calculation steps
 - Some examples

Unit 1: Probability models and axioms Lecture outline

- Sample space
- Probability laws
 - Axioms
 - Some properties
- Examples
 - Discrete
 - Continuous
- Interpretations of probabilities

Probability calculation steps (sequence of 4 steps)

1. Specify the sample space
2. Specify the probability law
3. Identify an event of interest
4. Calculate...

Practice Problem 3 – *do it in pairs*

Hasan has a peculiar pair of four-sided dice. When he rolls the dice, the probability of getting any particular outcome is proportional to the sum of the results of the two individual dice.

- (a) What is the probability of the sum being even?
- (b) What is the probability of Hasan rolling a 2 and a 3, in any order?

Practice Problem 3 - Solution

Hasan has a peculiar pair of four-sided dice. When he rolls the dice, the probability of getting any particular outcome is proportional to the sum of the results of the two individual dice.

- (a) What is the probability of the sum being even?
- (b) What is the probability of Hasan rolling a 2 and a 3, in any order?

Die 1	Die 2	Sum	P(Sum)
1	1	2	2p
1	2	3	3p
1	3	4	4p
1	4	5	5p
2	1	3	3p
2	2	4	4p
2	3	5	5p
2	4	6	6p
3	1	4	4p
3	2	5	5p
3	3	6	6p
3	4	7	7p
4	1	5	5p
4	2	6	6p
4	3	7	7p
4	4	8	8p
		Total	80p

(a)

$$P(\text{Even sum}) = 2p + 4p + 4p + 6p + 4p + 6p + 6p + 8p = 40p = \boxed{1/2}$$

(b)

$$P(\text{Rolling a 2 and a 3}) = P(2, 3) + P(3, 2) = 5p + 5p = 10p = \boxed{1/8}$$

$$P(\text{All outcomes}) = 80p \text{ (Total from the table)}$$

and therefore

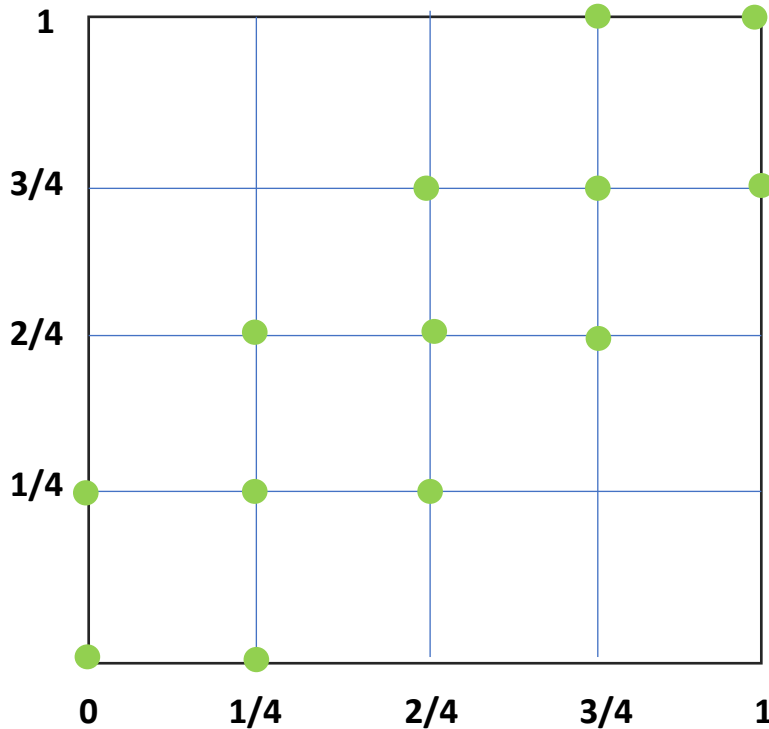
$$p = \frac{1}{80}$$

Practice Problem 4

“S” and “A” have a meeting at a given time and each will arrive at the meeting place with a delay between 0 and 1 hour – **all pairs of delay being equally likely**. The first to arrive will wait for 15 minutes and will leave if the other has not yet arrived. **What is the probability that they will meet?**

Two Cases of equally likely times –

- ➡ A) Arrival times are Discrete (equally likely) for both ‘S’ & ‘A’ - in steps of 15 mins ($t = 0, 1/4, 2/4, 3/4, 1$)
- B) Arrival times are Continuous



$$P(\text{Meeting}) = 13/25$$

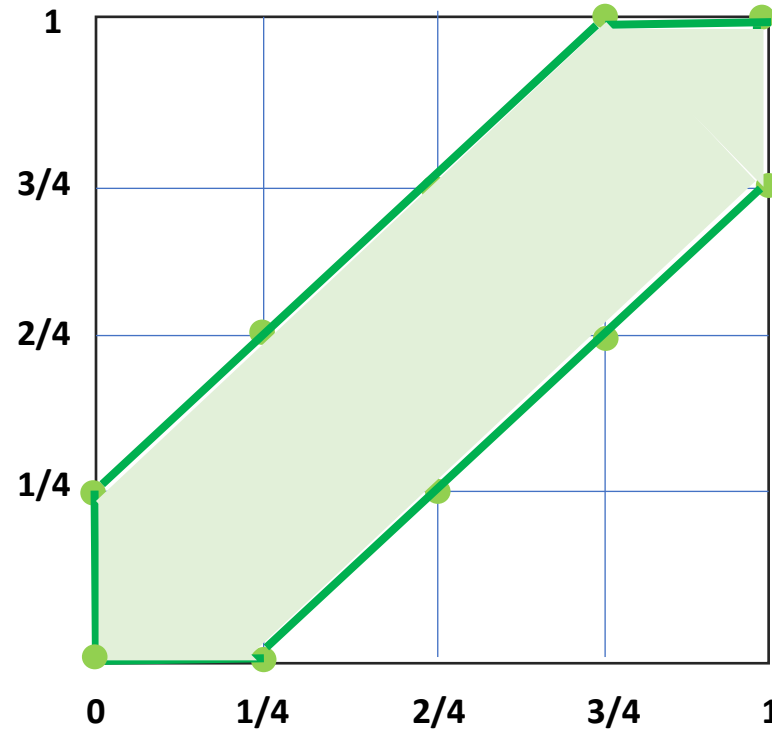
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➡ B) Arrival times are Continuous



$$\begin{aligned} P(\text{Meeting}) &= 1 - [2 \times (1/2 \times 3/4 \times 3/4)] \\ &= 1 - 9/16 \\ &= 7/16 \end{aligned}$$

Conditioning & Bayes' rule

- Let's say that you roll two dice and their sum comes out to be S .
 - ❖ If $S = 9$, what can we say about the first and second roll?
 - ✓ What is the probability that first roll is 1 or 2?

New knowledge should be used to modify our original beliefs.



Revised probabilities are called the conditional probabilities.